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Recyclens : AI-Based General Waste & E-Waste Intelligence System

A PROJECT REPORT

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

St. XAVIER'S CATHOLIC COLLEGE OF ENGINEERING

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APRIL 2026

BONAFIDE CETIFICATE

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

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LIST OF ABBREVIATION

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
IoT	Internet of Things
CNN	Convolutional Neural Network
ANN	Artificial Neural Network
YOLO	You Only Look Once
YOLOv5	You Only Look Once Version 5
YOLOv6	You Only Look Once Version 6
YOLOv8	You Only Look Once Version 8
YOLOv26n	You Only Look Once Version 26 Nano
mAP	Mean Average Precision
mAP@0.5	Mean Average Precision at IoU 0.5
mAP@0.5:0.95	Mean Average Precision (0.5 to 0.95 IoU)
GPU	Graphics Processing Unit
CPU	Central Processing Unit
SGD	Stochastic Gradient Descent
IEEE	Institute of Electrical and Electronics Engineers
CVPR	Conference on Computer Vision and Pattern Recognition
IJERT	International Journal of Engineering Research & Technology

Abbreviation	Full Form
IJIRSET	International Journal of Innovative Research in Science, Engineering and Technology
DFL	Distribution Focal Loss
lr	Learning Rate

ABSTRACT

Improper disposal of waste results in various issues such as environmental pollution and negative health effects. Most systems available today require manual sorting, which tends to be tedious, time-consuming, and inaccurate.

This project proposes Recyclens – an AI-based smart waste and e-waste intelligence system. This intelligent system makes use of advanced technologies such as artificial intelligence, machine learning, and internet of things in order to identify, classify, and segregate the different forms of waste. An artificial neural network based on YOLO detection algorithm is used in detecting various forms of waste such as non-biodegradable, biodegradable, and e-waste.

Recyclens incorporates both the hardware system, which includes sensors, cameras, conveyor belts, motors, among others, with the software system made up of a web application and a database.

In conclusion, the proposed system will save time since it reduces the involvement of human beings in sorting out waste.

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

The management of waste has become an issue of significant importance just within the last few years as population growth continues to grow, and more people are moving into urban areas. A consequence of poorly managed waste is the ability for waste to contaminate our land and water resources; further, the health and safety of people and animals will be greatly impacted by improperly handled waste.

Many locations continue to sort waste manually. The existing processes are often very slow (when compared to automating the processes) and require lots of manual labor; thus, errors are common.

To help solve these issues, an information technology company has developed a new solution referred to as Recyclens: Smart Waste and E-waste System with AI/ML-based technology. Recyclens will enable the automated detection, classification, and segregation of waste using the latest technologies, including Artificial Intelligence, Machine Learning, and Internet of Things technologies.

The physical identification of the various waste types (via a deep learning model) will be done by means of the YOLO algorithm; the waste types will include the classification of waste into three distinct categories: biodegradable, non-biodegradable, and e-waste. If waste is properly classified, then it can help to enhance overall recycling efficiencies and reduce the environmental impact caused by waste.

Recyclens uses integrated hardware and software components in order to perform its functions. The physical pieces (i.e. motors, conveyor systems, sensors, and cameras) associated with the hardware will detect the waste and capture images of the waste. The physical classification, detection, and segregation of waste will be done through the use of integrated hardware. Web-based technology will be used to manage data and monitor activity related to the waste being managed. Therefore, by using the Recyclens solution, the amount of manual work associated with waste management will be significantly reduced while improving the accuracy of the data captured regarding how the waste materials are being handled.

1.1.1 Importance of Smart Waste Management System

Maintaining a clean and healthy environment requires the proper management of waste. Wastes that are not sorted properly result in many recyclable materials being lost and/or contaminating the area with dangerous materials.

Automated systems for managing waste will help with this issue through automation and monitor waste on a real-time basis; This reduces the need to manually sort through the waste, improves sorting accuracy, and increases recycling rates. These types of systems are especially beneficial for managing hazardous materials, including electronic equipment (e-waste).

Adopting intelligent waste management solutions allows society to achieve a more environmentally friendly and sustainable future. recycled.

1.1.2 Role of Technology of Waste Management System

The Modern technology is enhancing and increasing the effectiveness of waste management systems to a much greater level than ever before. With new technology (e.g., Artificial Intelligence, IoT, and imaging technologies), it has become possible to automate the process of classifying and monitoring waste materials.

In this project, sensors are being used to detect waste, while images are captured using cameras. The captured images are then processed through a Deep Learning model to determine what kind of waste is present. The system also has mechanical components that will separate the waste according to the classification provided by the AI model.

By utilizing these modern methods, there is an increase in both speed and accuracy when compared to the traditional way of doing things.

1.1.3 Limitation of Existing System

Many current waste recycling systems rely on manual labor, which slows down the process and increases the possibility of human error and difficulty for workers in identifying different waste types, especially with large quantities.

Because of a lack of live monitoring and recording when it comes to handling recyclable materials, it is very difficult to keep recyclable materials clean so society can be clean. In addition, most current recycling systems are very expensive and also require continuous manual intervention.

Because of the many shortcomings found in these current systems, an intelligent automated recycling system should be developed that can provide efficient management of recyclable material. of waste.

1.2 PROBLEM STATEMENT

With regards to the issues surrounding tracking and managing recycling materials, not having the appropriate trackable systems in place will diminish the quality of recycling materials as well as not being able to provide a clean environment.

If recycling materials are not managed in an optimal manner then those recycling materials will likely become contaminated which will then make the reverse the efficacy of the recycling process.

In addition, the current recycling systems are typically costly and highly dependent on manual labor. As such the cost of the recycling process increases and therefore the efficacy of the overall recycling process may not be very reliable.

A solution to these problems is that we need a more intelligent (smarter) recycling system. This type of smarter recycling system will manage recycling materials, reduce the need for human support, and improve overall efficiency levels.

1.3 OBJECTIVES OF THE STUDY

The purpose of our project is to develop an AI-based Smart Waste Management System which will automatically identify and classify the various types of waste generated in our society and helps in recyclability

The objectives of this project are to decrease the amount of manual labor involved in handling wastes, improve the accuracy of waste classification, and increase the efficiency of recycling. This project will also create a proper method for disposing of hazardous electronic wastes (e-wastes) and will ensure that such disposal is done in an environmentally safe manner. Our intention is therefore to deliver a long-term reliable and practical solution to today's waste management problems.

1.4 SCOPE OF THE PROJECT

This system offers applications across several environments: Smart Cities, public spaces/areas, Industries, Recycling Facilities, etc. As such, it will enable improved separation of waste and lessen human labor requirements associated with this task.

Also, the system will provide valuable information for monitoring / assessment purposes helping local authorities improve their decision making concerning Waste Management. Additionally, this product has a design that can easily be expanded upon for larger uses in the future.

1.5 METHODOLOGY OVERVIEW

This process works in sequence. Detecting waste occurs first through sensors, then getting images of the waste via cameras. These images are run through an image recognition system (YOLO based AI algorithm) for identifying the waste type (sample).

Then, the system will use motors and conveyor systems to sort the waste as it is identified by the AI system. Simultaneously, this information is being stored in a database and monitored via an online application.

Our solution is designed to provide an efficient, automated method of managing waste while requiring minimal human .

1.6 SIGNIFICANCE OF THE STUDY

This solution presents itself as a clever and effective means of tackling the ever-growing challenge of waste management by relying on modern technology to automate the process of sorting and separation of waste, which ultimately contributes to the reduction of garbage overall.

What is more important, such a technological innovation can also make its contribution to pollution reduction since, while encouraging recycling and contributing to efficient waste management, it minimizes the volume of garbage that would otherwise end up in landfills.

Finally, this idea may prove helpful for governments, businesses, and environmental protection organizations working toward creating a cleaner and healthier environment.

1.7 ORGAINATION OF THE REPORT

This report is structured in a such a clear and simple way to help and Understand this project step by step

Chapter 2: Literature Review

Literature Review provides the existing waste management systems and technologies.

It discusses how they work, along with their limitations and the gaps that need improvement.

Chapter 3: Proposed System

The proposed system tells how the system is designed and covers the overall architecture of our solution and different modules of our system, and how the system works in whole.

It also describes the tools, technologies, and the development process of the solution used to create the project.

Chapter 4: System Testing

The system testing focus on how the system is built and system is tested and evaluates.

Chapter 5: Results and Discussion

This chapter presents how the system is implemented and evaluates its performance. It also includes sample outputs to show how well the system works.

Chapter 6: Conclusion and Future Enhancements

The final chapter of the project summarizes the results of the project and suggests possible improvements and future developments to make the system even better.

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CHAPTER 2

LITERATURE REVIEW

A review of literature pertaining to intelligent waste management is provided in this chapter. It discusses some key areas of application, including; the use of artificial intelligence for waste classification, such as using deep learning algorithms (e.g., YOLO), implementing IoT-based monitoring systems, and managing electronic waste.

The goal of this review is to provide an overview of the current status of existing systems, identify existing limitations of those systems, and identify existing research gaps, in order to assist with developing a more effective and efficient AI based Waste Management System.

2.1 Smart Waste Management System

Waste management has become a major global concern due to rapid urban growth and increasing population. Traditional methods of waste collection and handling are often slow, inefficient, and contribute to environmental pollution.

To solve the increasing issues in the management of waste, intelligent waste management systems have come into existence in the last few years. These waste management techniques use advanced technology like sensor-based waste management systems to improve efficiency and reliability.

The majority of the current waste management systems include smart bins using IoT that monitor their capacity and allow for an optimized schedule for waste collection purposes. This makes the process of waste collection cheaper as well as efficient. Yet, the current waste management systems usually concentrate on waste collection without considering proper waste segregation.

The lack of segregation in waste leads to inefficient recycling processes as well as large quantities of waste going into the landfills, hence affecting the environment. Therefore, the current proposal emphasizes on waste detection and segregation in order to improve efficiency.

2.2 Artificial Intelligence in Waste Classification

Artificial Intelligence, is beginning to take over the recycling and waste management industry by turning to the use of Machine Learning Algorithms and Deep Learning Algorithms to assist in the classification of different kinds of waste such as plastics, paper, metal, and organic material.

By doing so, they eliminate the need of having humans sort through items manually and greatly speed up the waste management process. Studies show that the integration of AI image classification models into the sorting process improves both the efficiency and accuracy of waste sorting.

Out of all the available types of Machine Learning Techniques, CNN’s are one of the most commonly used types of algorithms due to their superior ability to learn complex features from large datasets and classify objects with high levels of accuracy. Nonetheless, several significant challenges remain with implementing CNN's for the sorting of waste; these include inconsistent training data, varying lighting and shadows during Image Capture, and objects partially obscured by other objects.

To solve these problems, the proposed system integrates advanced Deep Learning Techniques with a Real-Time Image Processing Component, thus increasing the overall performance of the system and the reliability and effectiveness of the system with respect to real-world use.

2.3 Object Detection with YOLO

One of the most commonly used algorithms for object detection in computer vision applications is YOLO, which stands for "You Only Look Once". One reason why YOLO is so widely used is that it can quickly process images or video frames to find objects within them.

YOLO is different from other algorithms because it does not break an image into smaller sections or analyze an image in multiple stages; it performs a single analysis of the entire image without sacrificing accuracy, which allows for the use of YOLO in applications requiring real-time performance.

Different variants of YOLO exist, including YOLOv2 - YOLOv4. Each variant includes improved performance in terms of accuracy and speed as well as improved reliability and new features that previous versions lacked.

One advantage of using YOLO is its ability to detect multiple objects at once, making it a good candidate for use in applications involving automatic waste separation, identifying various classes of waste products in a single image or video frame

The model used for this project will utilize the YOLO algorithm to capture multiple different types of waste including biodegradable waste, non-biodegradable waste, recyclable materials, and hazardous electronic waste. This will improve the time and accuracy with which waste is sorted from one another.

2.4 Waste Monitoring Using IoT and Sensors

The real-time tracking and collection of data through the internet of things (IoT) are made possible through the use of IoT. Ultrasonic sensors, infrared sensors, and cameras play a major role in collecting data on how much waste is in a bin and what type of image the waste looks like.

The collection of waste data helps to inform how much waste is in each bin, providing information that facilitates better ways of managing garbage collection and waste disposal. The use of IoT systems creates an opportunity for communication between hardware and software, allowing for the processing and analysis of data collected from the hardware.

This will facilitate functions such as monitoring the physical status of the bin, improving the route the garbage truck takes to pick up the garbage, and avoiding overflowing the bin. Unfortunately, the majority of existing IoT systems do not possess intelligent decision-making capabilities.

The current gap in intelligent decision-making capabilities will be solved with the proposed integration of IoT with artificial intelligence (AI) to create a smarter and more automated system for the management of garbage, and ultimately for saving the environment.

2.5 Management Systems for E-Waste

Electronic waste is the fastest growing form of waste generated from increasing use of electronic devices. Improper disposal of e-waste can lead to major environmental problems due to the presence of toxic chemicals like lead, mercury and cadmium.

The threat posed by these hazardous substances to land and water quality, as well as to air, means that it is crucial that e-waste is handled properly.

Most e-waste management systems today focus primarily on collecting and recycling e-waste but do little to automate the identification and classification of the various components. As a result, the industry needs smarter e-waste management solutions which can identify the hazardous nature of certain components, like batteries, and recommend safe disposal of them.

2.6 Summary of Finding

Despite making significant progress in smart waste management, based on a review of the current literature we still see that most solutions do not fully optimize their systems. Many of the current solutions are only partially automated and do not effectively integrate their hardware and software components resulting in limitations in their efficiency and overall performance.

To address these limitations, we have proposed a system called Recyclers, which is an integrated system that combines several technologies to form one integrated solution to help create a more efficient waste management system.

This system utilizes YOLO for "Real Time" Waste Identification, AI will allow for accurate identification of multiple types of waste, and therefore allow for automated assorting. The Recyclers system will supply analytics and reporting features for e-waste as well as providing analysis for safe handling of the e-waste.

This integrated approach improves efficiency, reduces environmental impact, and promotes sustainable waste management practices.

Table 2.1: Summary of the magazine

S. No.	Author & Year	Title	Publisher	Methodology	Advantages	Disadvantages
1	Redmon et al., 2016	YOLO: Unified Real-Time Object Detection	IEEE CVPR	CNN, Single-stage detection	High speed, real-time detection	Lower accuracy for small objects
2	Redmon & Farhadi, 2018	YOLOv3: Incremental Improve me nt	arXiv	Deep CNN (Darknet-53)	Better accuracy, multi-scale detection	Computational complexity
3	Bochkovskiy et al., 2020	YOLOv4: Optimal Speed & Accuracy	arXiv	CSPDarknet, CNN	Improved speed & accuracy	Requires high GPU power

4	Pardini et al., 2020	Smart Waste Managemen t Solution	Sensors (MDPI)	IoT Sensors, Smart bins	Efficient waste monitorin g	No automatic segregation
5	Nowakowski et al., 2020	Deep Learnin g in E-Waste Collection	Elsevier	AI, Image Classification	Improves collection planning	Limited real-time detection
6	Rutqvist et al., 2019	Automated ML for Waste Systems	IEEE Transacti ons	Machine Learning Models	Reduces manual effort	Accuracy depends on dataset
7	Özkan et al., 2015	Plastic Waste Classificatio n	Elsevier	Classification Algorithms	Improves recycling efficiency	Limited to plastics only
8	Omran et al., 2021	AI in Waste Managemen t (Review)	Springer	Survey of AI Techniques	Compreh ensive analysis	Lacks implementati on model
9	Chen et al., 2018	Tinier-YOLO for Embedded Systems	IEEE Conferen ce	Lightweight CNN	Suitable for low-power devices	Reduced accuracy

10	Smart Bin Research (Various)	IoT-Based Waste Monitoring	IEEE / Elsevier	Sensors, Cloud systems	Real-time monitoring	No intelligent classification
11	Zhang et al., 2021	Deep Learning-Based Waste Classification	IEEE Access	CNN, Image Processing	High classification accuracy	Requires large dataset
12	Kumar & Singh, 2022	IoT-Based Smart Waste Segregation System	IJERT	IoT Sensors, Automation	Real-time monitoring & segregation	Hardware cost is high
13	Li et al., 2020	Automated Waste Sorting System	Elsevier	Machine Learning, Robotics	Reduces manual effort	Complex system design
14	Sharma et al., 2023	AI-Based Waste Detection Using CNN	Springer	Deep Learning Models	Fast and efficient detection	Performance depends on training

15	Wang et al., 2021	Intelligent Garbage Classification System	IEEE Transactions	CNN, Transfer Learning	Improved accuracy	High computational cost
16	Patel et al., 2022	Smart E-Waste Management System	IJIRSET	IoT, Cloud Computing	Efficient e-waste tracking	Limited scalability
17	Singh et al., 2021	Real-Time Waste Detection Using YOLO	IEEE Conference	YOLO Algorithm	Real-time detection	Struggles with small objects
18	Chen et al., 2022	AI-Based Recycling System	Elsevier	Deep Learning, Automation	Improves recycling rate	Expensive implementation
19	Gupta et al., 2020	Waste Segregation Using Image Processing	Springer	Image Processing Techniques	Simple implementation	Lower accuracy than AI
20	Rahman et al., 2023	Smart Waste Monitoring Using IoT	IEEE Access	IoT, Sensors, Cloud	Efficient data collection	Depends on internet connectivity

CHAPTER 3
PROPOSED SYSTEM

3.1 INTRODUCTION

This chapter presents the **"Recyclens: AI-Powered Smart Waste and E-Waste Intelligence System"** system. The goal of developing this proposed solution is to provide smart waste management based on artificial intelligence, deep learning, and innovative hardware.

The main objective of our proposed system is to enhance the waste segregation process by using automatic detection and identification algorithms for different types of waste such as e-waste, biodegradable, and non-biodegradable items. Therefore, we chose a combination of the YOLO-based deep learning algorithm with a web application and the associated hardware to accomplish this goal.

Consequently, it is possible to conclude that the Proposals will optimize the waste segregation process and encourage environmental sustainability.

3.2 OVERALL SYSTEM OVERVIEW

Recyclens is an innovative solution for improving waste management by combining both software and hardware to develop a automated solution to sort and recycle materials. The solution has been designed to automatically identify waste material, take a picture of it, and analyze it with AI algorithms to identify the specific type of material.

There are several major parts of the system:

1) The sensor and camera module detects the presence and takes an image of the waste for further analysis.

2) The AI detection and classification module utilizes deep learning technology to classify the material type.

3) The automatic segregation module uses the classification of the waste material to direct it to the appropriate bin.

The smart analysis module provides information regarding the recyclability of the materials and their environmental impact. The database and reporting module stores the information that has been collected and generates reports for monitoring and analyzing data.

As you can see, the Recyclens Recycling System works in a continuous loop, with little or no need for human intervention, making the entire process of waste management smooth and effective.

3.3 SYSTEM ARCHITECTURE

Recyclens is an innovative solution for improving waste management by combining both software and hardware to develop a more automated method to sort and recycle materials. The system has been designed to automatically identify waste material, take a picture of it, and analyze it with AI algorithms to identify the specific type of material.

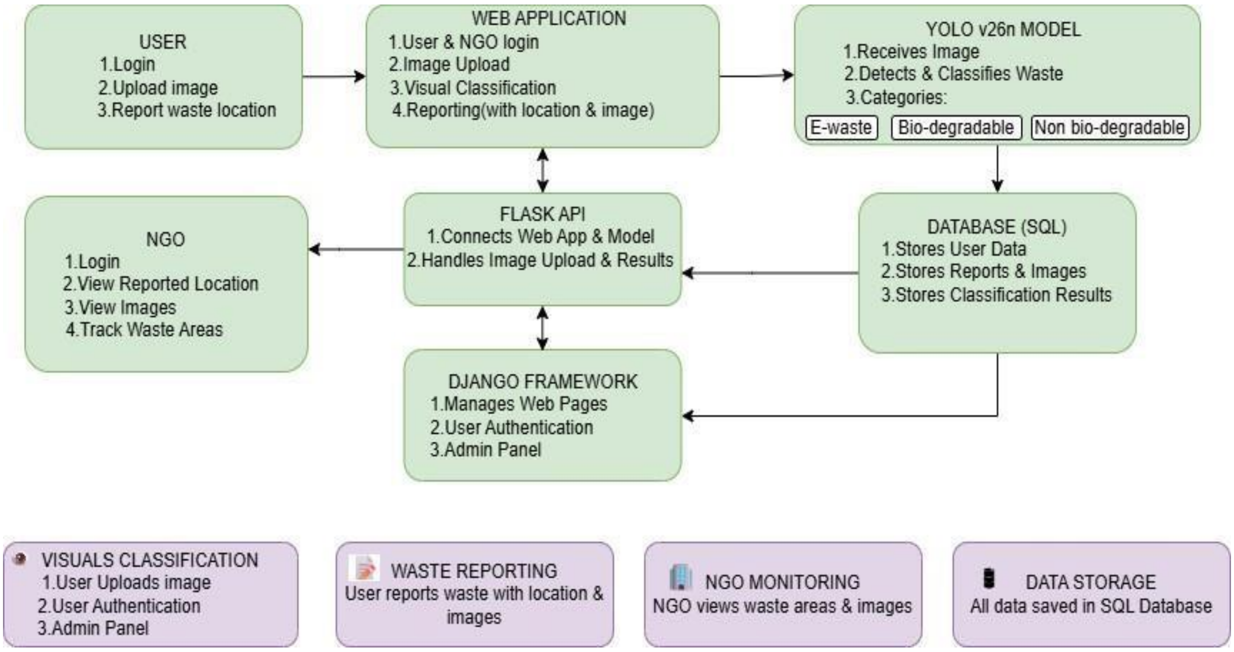


Figure 3.1:System Architecture diagram

There are several major parts of the system:

1) The sensor and camera module detects the presence and takes an image of the waste for further analysis.

2) The AI detection and classification module utilizes deep learning technology to classify the material type.

3) The automatic segregation module uses the classification of the waste material to direct it to the appropriate bin.

The smart analysis module provides information regarding the recyclability of the materials and their environmental impact. 5) The database and reporting module stores the information that has been collected and generates reports for monitoring and analyzing data.

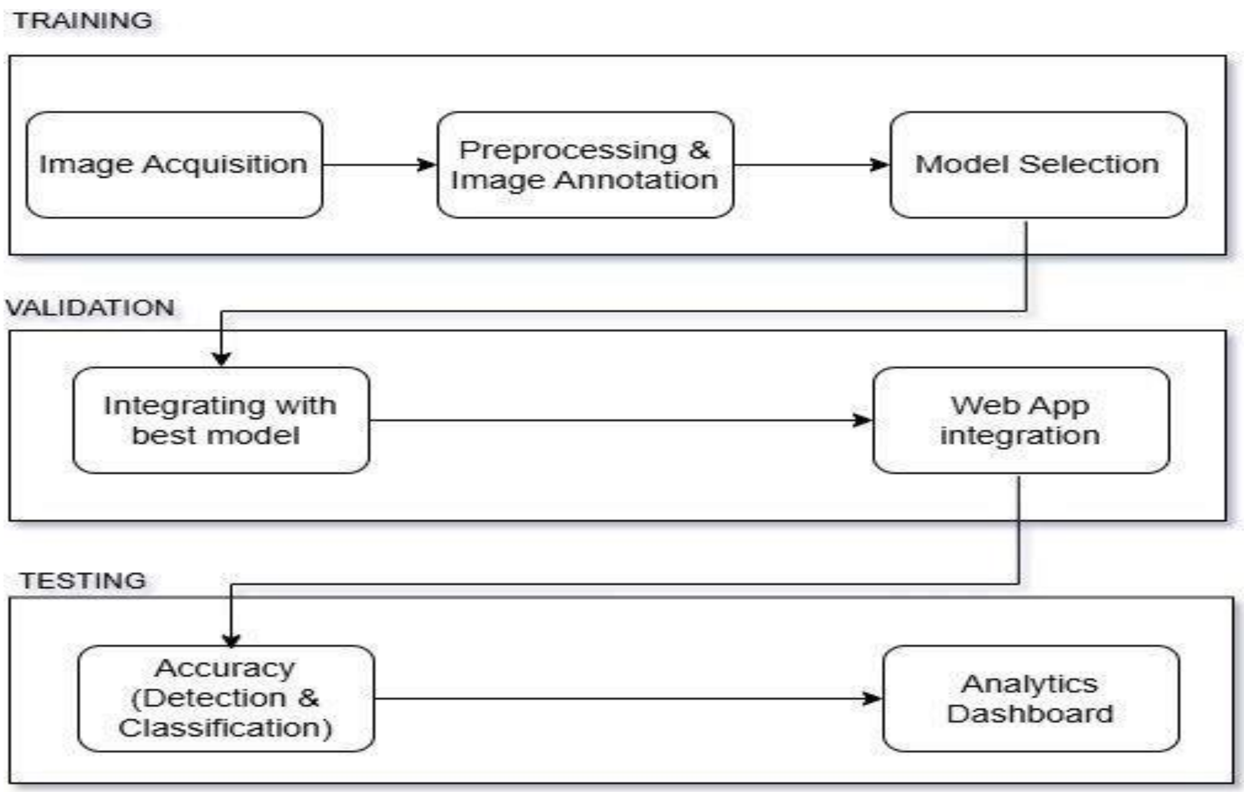


Figure 3.2 : System flow

As you can see, the Recyclens Recycling System works in a continuous loop, with little or no need for human intervention, making the entire process of waste management smooth and effective.

3.1 SYSTEM MODULE

The Recyclens system is divided into several functional modules

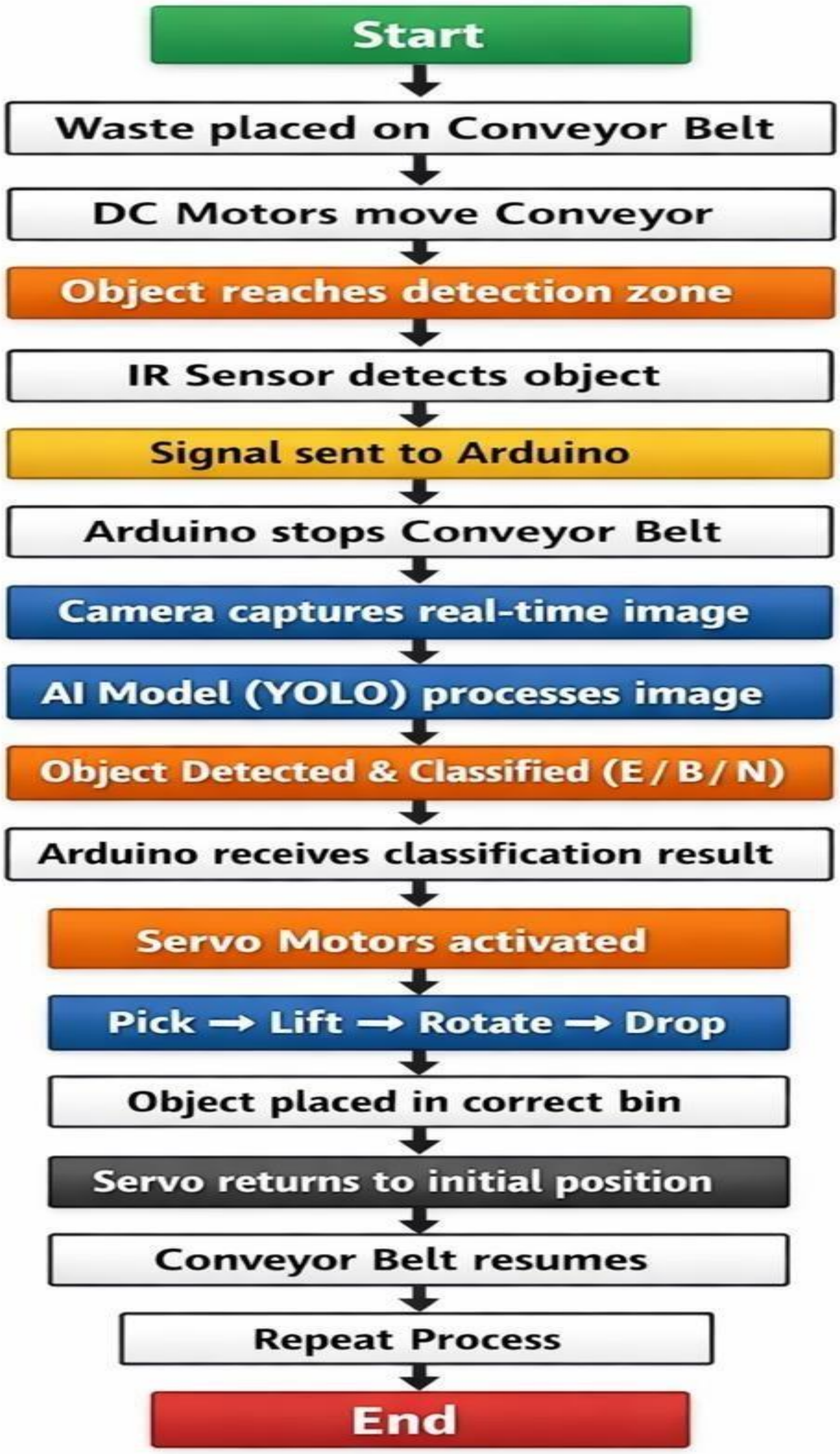


Figure 3.3: System Module diagram

3.1.1 Sensor and camera module

Uses sensors to identify objects and capture images of the object for analysis; serves as input to the system

3.1.2 AI detection and classification module

Uses deep learning analysis of the images captured using the AI detection module by identifying the type of waste; results in high accuracy and efficiency.

3.1.3 Automatic segregation module

Based on the results from the classification module, sends signals to the mechanical components for directing the waste to the appropriate bins; eliminates human manual contamination (e.g. waste that is placed in the wrong bin) and increases efficiency

3.1.4 Smart analysis module

Provides additional analysis capabilities, such as evaluating whether the type of waste is recyclable, predicting if it contains hazardous materials (e.g. battery), estimating the amount of material that can be recovered, and giving insight into the environmental impact of waste; assists with decision making and monitoring.

3.1.5 Database and Reporting module

Stores all of the data produced by the system (e.g. waste classification results), generates reports which can be used to monitor the performance of the system and provide information to improve waste management practices.

3.1 SYSTEM WORKFLOW

The workflow of the **Recyclens: AI-Based Smart Waste and E-Waste Intelligence System** describes how waste is detected, classified, and managed in a step-by-step process.

The process begins when the camera captures an image of the waste material. This image is then sent to the software system for further processing. Once received, the AI model analyzes the image and performs detection and classification of the waste.

After classification, the system determines the type of waste**.** If the waste falls under general waste, it is further divided into degradable and non-degradable categories. If the system identifies the waste as e-waste**,** it is further classified into subcategories such as metal and hazardous materials**.**

In cases where the system is unable to confidently identify the waste, it is labeled as unknown, and the process is directed to manual checking to ensure accuracy.

Once the waste type is confirmed, the system sends a signal to the hardware module, which controls the physical segregation process. The waste is then automatically directed to the appropriate bin based on its category.

After segregation, all relevant data, including classification results and waste details, are stored in the database. This stored information is later used for analysis and report generation, helping in monitoring waste patterns and improving management strategies.

Overall, the workflow operates in a continuous and automated manner, ensuring efficient waste handling, accurate classification, and minimal human usage.

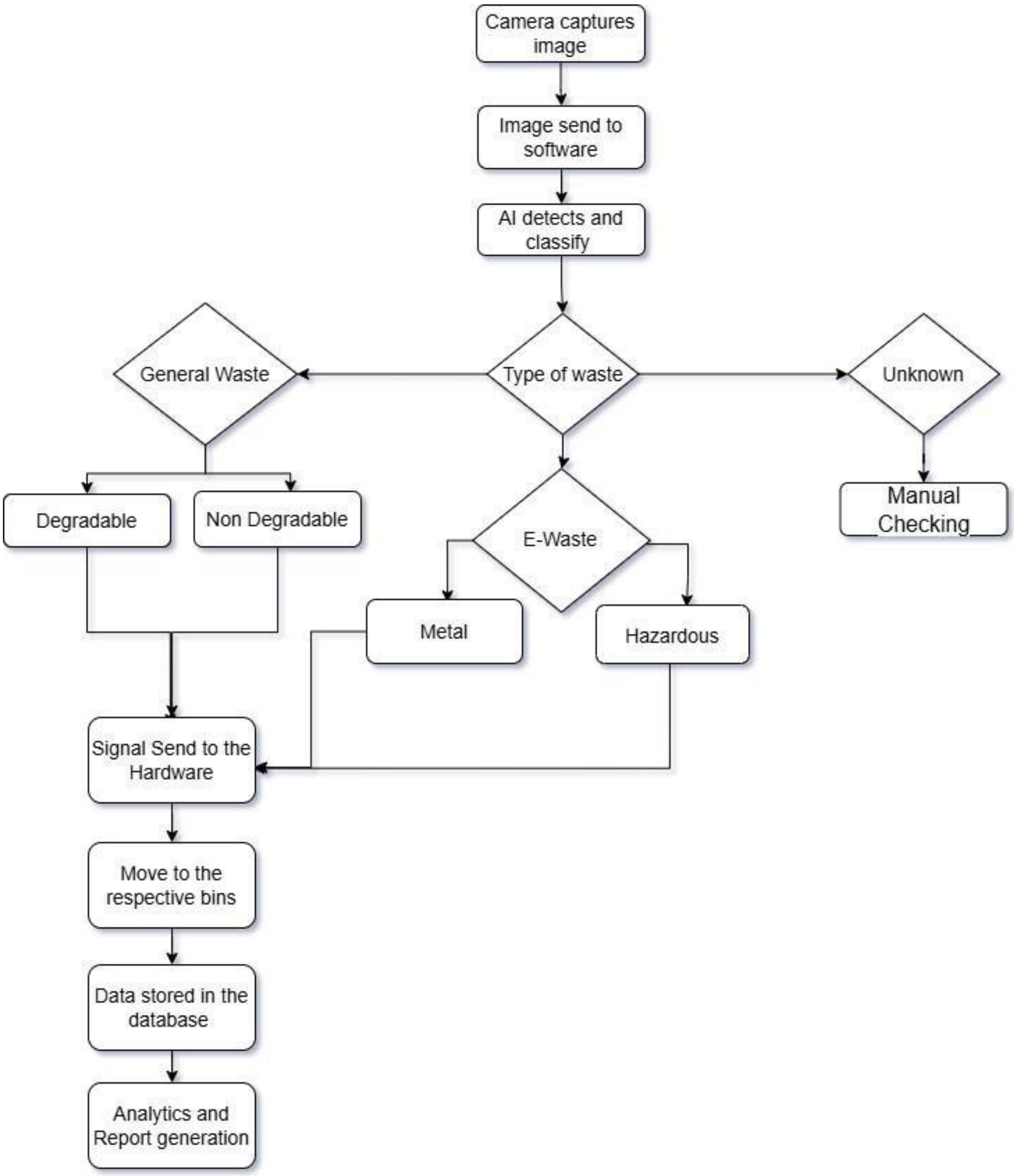


Figure 3.4: System Workflow diagram

3.2 ADVANTAGES OF THE SUGGESTED SYSTEM

The Recyclens system has a number of strengths including:

- Automation of waste identification and sorting
- Enhanced accuracy with AI classification techniques
- Minimal labor involved
- Real-time monitoring and reporting
- Safe handling of hazardous e-waste
- Scalable and efficient system design

The system provides a smart, efficient, and sustainable solution for modern waste management challenges.

CHAPTER 4
SYSTEM TESTING

4.1 INTRODUCTION TO TESTING

System testing is an essential component in the development of **Recyclens:** AI-Driven Intelligent waste and E-Waste System, which verifies that the system is functioning as intended; complies with all Business Requirements; and delivers predictable, affordable performance throughout all potential environmental variations.

Testing is required to identify, document and correct any potential defects or non-conformance of the system to prevent issues from having an impact on how smoothly or reliably the system operates. In addition, testing is essential to evaluate the overall performance of the system and confirm that the Waste Detection Process, Waste Image Processing, AI-Based Classification, Automated Waste Segregation Mechanism and all Database Operations function properly.

Testing involves reviewing each function of the system to ensure it works as intended and the quality of every component meets expectation. Therefore, the purpose of testing is to make it possible for the entire system to function efficiently, with consistent quality, prior to making it available for actual use.

In addition, testing verifies the proper interaction between hardware components (sensors, camera, motors) and software modules (AI model, web application, and database).

The testing activities are divided into:

- Module Testing – Testing individual components independently
- Integration Testing – Testing the complete system workflow

4.2 MODULE TESTING

Module testing (Unit Testing) is performed to verify each module of the system independently. It ensures that each component functions correctly before integration.

The Recyclens system consists of the following modules:

- Sensor and Camera Module
- AI Detection & Classification Module
- Automatic Segregation Module
- Smart Analysis Module
- Database & Reporting Module

4.2.1 Module 1 – Sensor and Camera Module Testin

This module captures waste images and detects object presence using sensors.

Test Case 1: Object Detection by Sensor

Verify that the sensor detects the presence of waste accurately when placed near the input area.

Test Case 2: Image Capture Functionality

Verify that the camera captures clear images of waste materials upon detection.

Test Case 3: Data Transmission

Verify that captured images are successfully transmitted to the AI processing module without delay.

4.2.2 Module 2 – AI Detection & Classification Module Testing

This module processes images using deep learning algorithms (YOLO-based model) to classify waste.

Test Case 1: Waste Classification Accuracy

Verify that the system correctly classifies waste into:

- Degradable
- Non-degradable
- Recyclable metals
- Hazardous e-waste

Test Case 2: Model Response Time

Verify that classification results are generated within an acceptable time

Test Case 3: Handling Unknown Objects

Verify system behavior when unidentified or unclear waste is detected.

4.2.3 Module 3 – Automatic Segregation Module Testing

This module directs waste into appropriate bins based on classification.

Test Case 1: Bin Selection Mechanism

Verify that the correct bin is selected based on classification output.

Test Case 2: Mechanical Operation

Verify that actuators or motors function properly to direct waste into bins.

Test Case 3: Error Handling

Verify that the system response when mechanical failure occurs on it.

4.2.4 Module 4 – Smart Analysis Module Testing

This module shows the additional insights such as recyclability and environmental impact.

Test Case 1: Recyclability Evaluation

Checks wheather the system estimates recyclable content correctly.

Test Case 2: Hazard Detection

Checks for the detection of hazardous components like batteries or electronic parts.

Test Case 3: Environmental Impact Indicators

Verify that the system provides any meaningful environmental insights on it

.4.2.5 Module 5 – Database & Report Module Testing

The module stores data and generates reports.

Test Case 1: Data Storage

Verify that the classification results and waste details are stored and maintained properly

Test Case 2: Report Generation

Verify that wheather reports are generated accurately for analysis.

Test Case 3: Data Retrieval

Verify that stored data can be accessed efficiently.

Verify that the system provides meaningful environmental insights.

4.3 INTEGRATION TESTING

Integration tests validate how well integrated the pieces of the program are and confirm that they work properly as a whole. The Recyclens integration test focuses on how well the hardware and software work together.

Test Case 1: Waste Detection to Classification Workflow

- Place waste near senso
- Capture image using camera
- Process image using AI model
- Verify correct classification output

Test Case 2: Classification to Segregation Workflow

- Classify waste using AI model
- Send control signal to hardware
- Verify that waste is directed into the correct bin

Test Case 3: End-to-End System Workflow

- Detect waste
- Classify waste
- Segregate automatically
- Store data in database
- Generate report

Test Case 4: Smart Monitoring and Analysis

- Verify that system tracks waste data
- Check analysis results and environmental indicators
- Validate report accuracy

4.4 TESTING OUTCOME

The testing confirmed that:

- The AI model accurately classifies different types of waste
- Sensors and camera modules work efficiently
- Automatic segregation operates correctly
- All modules interact seamlessly
- The system improves waste management efficiency and reduces manual use.

The proposed solution which performs reliably and meets the requirements for smart waste management.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION

This chapter presents what actually happened when we trained and tested the four detection models we chose for comparison — YOLOv5, YOLOv6, YOLOv8, and YOLOv26n. All four were evaluated on the same dataset covering three waste types: electronic waste, biodegradable waste, and non-biodegradable waste. The goal was to find which model would serve Recyclens best in real operation, and the results pointed clearly toward YOLOv26n.

5.2 Experimental Setup

Rather than using a publicly available dataset, we built our own from waste samples collected across different real environments. This made the task harder for the models but also more relevant to actual deployment conditions. Every model trained on the same data with the same settings, so the comparison reflects genuine performance differences rather than setup advantages.

Data was divided into 70% training, 20% validation, and 10% for final testing. All experiments ran on an Intel Core i5 machine through Python using the Ultralytics library.

5.2 Experimental Setup

All tests were done on a specially created dataset that was classified based on electronic waste, biodegradable waste, and non-biodegradable waste collected from different environmental sites. In order to do an impartial experiment on the four models, the YOLO models were trained using the same dataset with similar hyperparameters.

As far as splitting the data is concerned, 70%, 20%, and 10% data were allocated for training, validating, and testing purposes, respectively.

All the models were evaluated using an Intel core i5 processor through Python programming language and the Ultralytics library. The hyperparameters used for the training process are as follows:

Table 5.1: Training Configuration Parameters

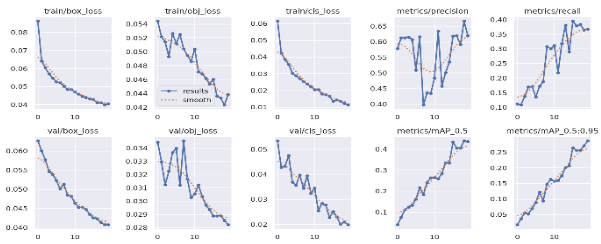
Parameter	Value
Batch Size	16
Epochs	50
Image Size	640 × 640
Optimizer	SGD (lr = 0.01)
Framework	Ultralytics (Python)
Models Used	YOLOv5, YOLOv6, YOLOv8, YOLOv26n
Dataset Classes	E-Waste, Biodegradable, Non-Biodegradable

5.3 Model Training Results

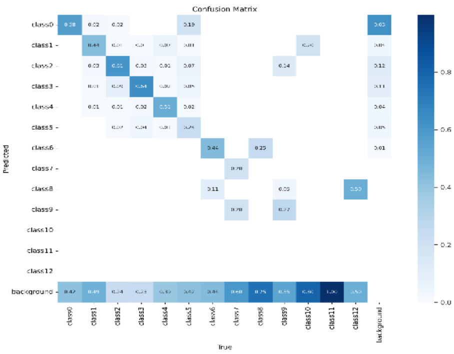
The training process involved the use of customized dataset for the training of each of the models independently. The evaluation was based on training curves that consisted of accuracy, precision, recall, and loss curves in order to understand the learning process of each model.

5.3.1 YOLOv5

YOLOv5 trained in a fairly predictable way — loss reduced steadily, precision and recall climbed gradually, and nothing unexpected happened during training. Where it showed weakness was in distinguishing waste from background elements. The confusion matrix revealed this happening more frequently than we wanted, which would cause real problems in an environment where background clutter is unavoidable.



(a) YOLOv5 Training Curves

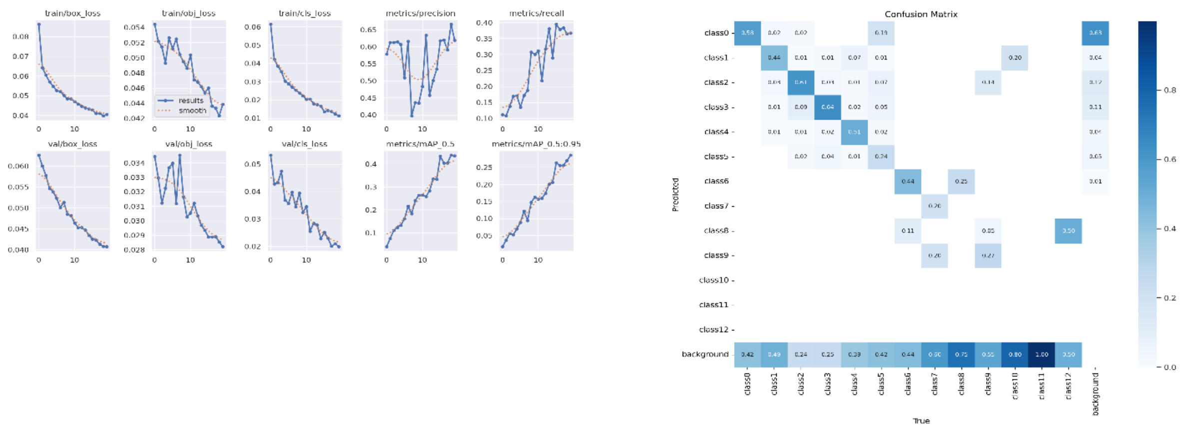


(b) YOLOv5 Detection Results

Figure 5.1: YOLOv5 Training Performance and Detection Output

5.3.2 YOLOv6

We genuinely expected more from YOLOv6 going in. It has a reputation for speed and efficiency, but neither translated well onto our dataset. Precision and recall both sat at very low levels throughout training and never improved meaningfully. Background interference was a persistent issue. When the final comparison numbers came in, YOLOv6 performed worst across almost every metric — which tells us it simply wasn't well-suited to this type of data.

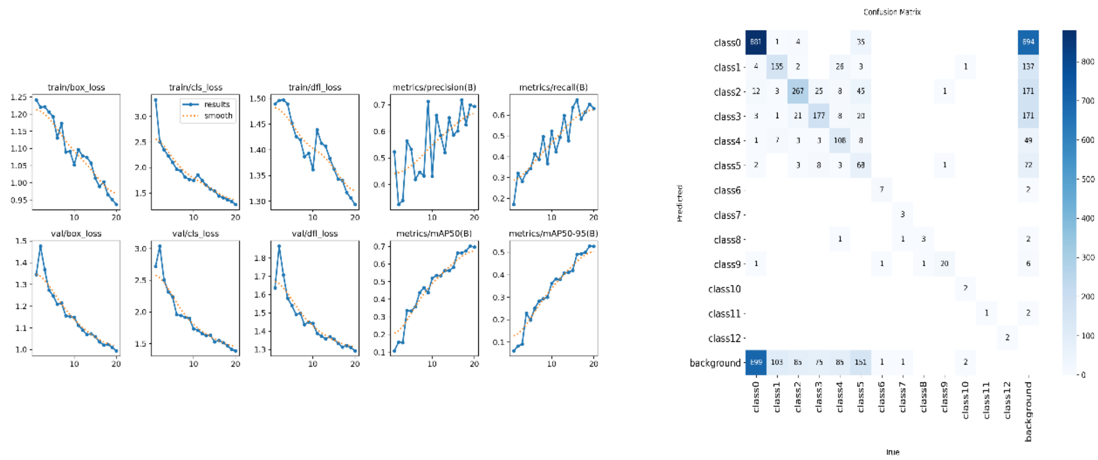


(a) YOLOv6 Training Curves (b) YOLOv6 Detection Results

Figure 5.2: YOLOv6 Training Performance and Detection Output

5.3.3 YOLOv8

YOLOv8 was the one that gave us the most to think about. Its architecture handles varied object types fairly well, and the loss metrics behaved healthily during training. It reached an mAP@0.5 of 0.70, which is a reasonable result. The problem was inconsistency — precision and recall kept fluctuating when they should have stabilised, and one waste class kept generating false positives. With more data or longer training it might have performed better. With what we had available, it fell short of what we needed.

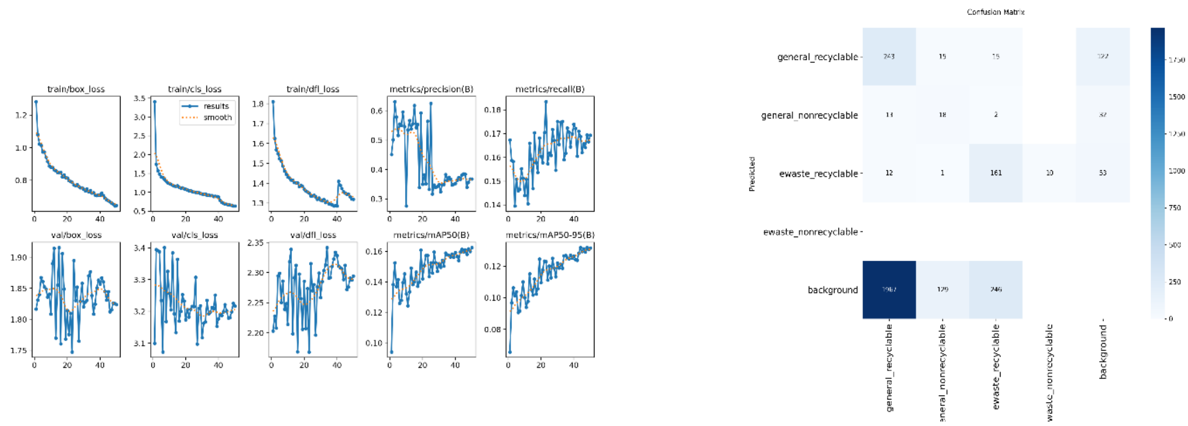


(a) YOLOv8 Training Curves (b) YOLOv8 Detection Results

Figure 5.3: YOLOv8 Training Performance and Detection Output

5.3.4 YOLOv26n

From fairly early in training, YOLOv26n behaved differently to the others. The learning curves were noticeably more stable, convergence happened faster, and classification accuracy held up consistently rather than fluctuating. For the main waste category it logged over 900 correct predictions, which reflects genuine learning rather than lucky guesses. It finished with an mAP@0.5 of 0.84 — the highest of the four — while also being the fastest and lightest model we tested. That combination is unusual and it made the selection decision straightforward.



(a) YOLOv26n Training Curves (b) YOLOv26n Detection Results

Figure 5.4: YOLOv26n Training Performance and Detection Output

5.4 Comparative Performance Analysis

As shown below, there is a comparison of all models on the basis of their performance on various parameters such as Precision, Recall, mAP@0.5, mAP@0.5:0.95, inference time and number of parameters. From the given results, it is apparent that YOLOv26n offers best performance by achieving an mAP@0.5 value of 0.84 which surpasses the performances offered by both

YOLOv5 and YOLOv6. Even though YOLOv8 has good precision, YOLOv26n offers better stability and lower inference time of only 9 ms/frame.

Model	**Precision**	**Recall**	**mAP@0.5**	**mAP@0.5:0.95**	**Inference (ms)**	**Params (M)**
YOLOv5	0.62	0.58	0.44	0.29	12	7.2
YOLOv6	0.12	0.09	0.07	0.04	14	10
YOLOv8	0.81	0.66	0.74	0.60	10	11
YOLOv26n*	0.85	0.72	0.84	0.65	9	6.5

Table 5.2: Comparative Performance Metrics Across All Evaluated Models

5.5 Model Selection for the Proposed System

From the results of the experiments presented in Table 5.2 and training analysis provided in Section 5.2, the object detection backbone chosen for use within the Recyclens AI-based intelligent waste management system is YOLOv26n. The reasoning behind choosing this algorithm lies in the following aspects:

- Lowest Rate of Misclassification: YOLOv26n exhibited the highest values for mAP@0.5 – 0.84 and mAP@0.5:0.95 – 0.65 among the studied algorithms, ensuring that waste is sorted with precision and avoiding classification errors between biodegradable and non-biodegradable items.
- Low Latency: With the latency rate of 9 ms per frame, YOLOv26n complies with the stringent requirements concerning the speed at which

waste can be analyzed while passing through the conveyor belt and smart bin systems.

- Smooth and Fast Convergence: Based on the training charts of YOLOv26n, it was found that this model exhibits fast convergence during training with low fluctuations.
- Convergent Training: The training charts of YOLOv26n demonstrate convergent learning with fast and stable curves, suggesting strong reliability and flexibility for learning and adapting to different classes of waste materials.
- Low Number of False Positives: The precision of YOLOv26n being 0.85 means that there is a relatively low number of false positives, which is essential in ensuring that recycled batches are not contaminated by other materials.
- Fewer Parameters: YOLOv26n operates with just 6.5M parameters, which makes it one of the most efficient object detection models in the industry.

5.6 Discussion

Going through all these results, a few things became clear that weren't obvious at the start. Model size doesn't reliably predict performance — YOLOv6 proved that. What seems to matter more is how well the architecture handles contextual variation, which in a waste-sorting environment means dealing with inconsistent backgrounds, varied lighting, and objects that don't always look the same twice. YOLOv26n handled that better than anything else we tested.

YOLOv8 remained a genuinely capable option and probably deserves more investigation with a larger dataset. But within the scope of this project and the resources we had, YOLOv26n was clearly the right backbone for Recyclens — practically deployable, accurate enough for real use, and stable enough to trust in a continuous operational environment.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

To conclude, it must be emphasized that it has been proved that designing a new product with the help of artificial intelligence, deep learning, and the Internet of Things is not particularly difficult. More importantly, there is no necessity for people to engage in sorting since the intelligent machine is unlikely to make any errors. The classification of waste is ensured due to the utilization of the detection model, which is built on the principles of YOLO.

The utilization of such components as the sensor, camera, conveyor belt, and servomotor enables sorting to occur automatically while the app controls this process. Overall, it must be noted that this platform will probably prove to be effective regarding waste management and recycling.

6.2 FUTURE ENHANCEMENTS

Nevertheless, even though this approach of tackling the problem is regarded as extremely efficient and it has been determined to be the most effective so far, there are still many areas that can be improved within this very system:

- Next Generation AI Solutions: Usage of next generation AI solutions in order to improve the classification accuracy.
- Mobile App Solution: Development of mobile apps to monitor pollution and control the process.
- Cloud Solution: Integration of cloud services and the use of them in order to increase the system's capacity for storing data and operate it from anywhere.

- IoT Solution: Installation of new IoT functionality and usage of various sensors for monitoring fill levels, triggering alerts and status changes.
- Alarm Solution: Notifications of all parties involved in cases when overfilling is predicted and potentially hazardous materials are found.
- Scaling: Improvement of software engineering methods used for scaling purposes and making this particular solution applicable to other businesses.
- Recommendation System: Creation of a recommendation system to recycle properly.